

Punam Adhikari

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PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in Artificial Intelligence, Machine Learning, Statistics, Data Engineering, and Research. Proficient in Python, SQL, and machine learning frameworks with strong data processing skills. Passionate about solving complex problems through data-driven solutions.

PROFESSIONAL EXPERIENCE

AI Research Engineer

Oct 2025 – Present

NSDevil

Onsite

- Fine-tuned Whisper, NeMo, and Conformer-based speech-to-text models for Korean (KSponSpeech, broadcast datasets), achieving under 10% WER
- Developed high-quality Korean text-to-speech systems using ESPnet
- Built an explainable AI-based Korean essay auto-scoring system using microservices and LLM-based rubric scoring
- Fine-tuned HyperCLOVAX (3B, 14B) models using SFT and DPO with LoRA/QLoRA across multiple academic domains
- Trained and deployed Transformer and multimodal models on remote GPU servers with CI/CD pipelines
- Designed and deployed a production-grade multimodal RAG system, improving domain-specific retrieval accuracy and reducing latency via optimized indexing and caching

Software Engineer-I

Apr 2025 – Oct 2025

Techkraft Inc. (Abacus Insights)

On-Site

- Applied NLP techniques for automated PDF data extraction in the US healthcare domain, improving accuracy and efficiency of information retrieval
- Designed, developed, and maintained ETL/data pipelines using Databricks, Snowflake, Python, PySpark and SQL
- Optimized data assets for quality, performance, and reliability in US healthcare data environments
- Monitored and troubleshoot pipelines to ensure uninterrupted data distribution
- Collaborated with cross-functional stakeholders to translate client requirements into scalable data solutions
- Worked in Agile environments with clear communication across engineering teams

AI/ML Research Intern

Nov 2024 – Apr 2025

Danson Solutions

Hybrid

- Engineered a code translation pipeline using Meta-LLaMA and GPT-3.5 Turbo to automate PHP-to-Java migration
- Fine-tuned and evaluated LLMs for code generation tasks using prompt engineering techniques
- Explored AI-driven code refactoring using embeddings and syntactic analysis
- Collaborated with a remote research team on multimodal datasets and documented experimental findings

EDUCATION

Bachelor's in Computer Engineering

May 2021 – June 2025

Pulchowk Engineering Campus

Kathmandu, Nepal

Higher Secondary (+2 Science)

2018 – 2020

Satyawati Secondary School

Tanahun, Nepal

PROJECTS

AI-Powered Online Exam Proctoring System

GitHub

Python, OpenCV, YOLO, Dlib, Scikit-learn, CNN

- Built a real-time proctoring system using webcam video and computer vision models
- Implemented 7 modules: person/object detection, face detection/verification, spoofing, gaze, and mouth analysis
- Used YOLO for detecting banned items and multiple persons; applied OpenCV SSD for face detection; verified users with Dlib's ResNet50 (128D embeddings)
- Detected spoofed faces using YCrCb + Luv histograms and ExtraTreesClassifier
- Tracked gaze via Eye Tracking Ratio (ETR) from 68 facial landmarks; detected speaking using Lip Aspect Ratio (LAR) and frame buffer logic
- Trained lightweight CNN for head pose estimation, achieving 10.79° MAE on the BIWI dataset

- Integrated all modules into one pipeline with live alerting, optimized for under 30 FPS on low-end machines

Flight Fare Prediction

GitHub

Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn

- Built a machine learning model to predict flight prices using a dataset of 11,000 rows and 11 features
- Performed extensive data cleansing on noisy, inconsistent, and mixed-type data using Pandas and NumPy
- Executed advanced preprocessing and feature engineering using scikit-learn pipelines, FeatureUnion, and ColumnTransformer
- Trained and deployed an XGBoost model as a Flask web app on Render

Multi-Class Brain Tumor Classification

GitHub

Python, TensorFlow, CNN, Transfer Learning

- Developed a deep learning model to classify brain tumors (glioma, meningioma, pituitary, no tumor) from 7,023 MRI images
- Built and compared a custom 10-layer CNN (97% accuracy) and a VGG-16 transfer learning model (91% accuracy) using TensorFlow/Keras
- Applied preprocessing including image resizing, normalization, and label encoding; trained with Softmax activation using Adam and SGD optimizers
- Used callbacks such as ModelCheckpoint and CSVLogger for efficient training monitoring
- Evaluated models using accuracy, F1-score, confusion matrix, and ROC curves; custom CNN outperformed VGG-16
- Deployed the CNN model through a Flutter mobile application integrated with a Django backend for real-time clinical inference

BankLoanAnalysis

GitHub

PySpark, Databricks, Power BI, Airflow, Azure Data Factory

- Designed and orchestrated ETL pipelines using Azure Data Factory to automate data ingestion from source systems into a cloud-based data lake
- Implemented DAGs in Apache Airflow to schedule and monitor PySpark jobs, ensuring reliable data transformation workflows
- Developed interactive dashboards in Power BI to visualize loan approval trends, customer segmentation, and credit risk patterns
- Utilized Databricks and PySpark to perform large-scale data cleaning, transformation, and feature engineering for model-ready datasets

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Unix/Linux Scripting, Java, Shell Scripting
- **Database and SQL:** PostgreSQL, MySQL, Oracle, MongoDB, Complex Queries
- **Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, EDA, Power BI, Tableau
- **Big Data and ETL Tools:** Airflow, Apache Spark, PySpark, Hadoop ETL pipelines, Databricks, Kafka, Apache Lambda, AWS S3, Amazon Redshift, Azure Data Factory, Azure Synapse Analytics, ADLS Gen2 Storage, Snowflake
- **Machine Learning and AI:** Scikit-Learn, TensorFlow, PyTorch, Keras, Model Validation, Deep Learning
- **Additional Skills:** Git, API Development using FastAPI, Docker

CERTIFICATIONS AND ACHIEVEMENTS

- **IGDTUW – Computer Vision and Deep Learning Summer Intern:** Nominated and selected to represent Pulchowk Campus (IOE) in an international academic internship focused on advanced topics in Computer Vision and Deep Learning; participated in collaborative research alongside students and faculty from leading institutions
- **Machine Learning Specialization:** Completed Andrew Ng's comprehensive machine learning specialization covering supervised and unsupervised learning algorithms, Coursera
- **HackerRank SQL Certificates:** Achieved Intermediate and Advanced SQL certifications demonstrating proficiency in complex queries
- **AWS Academy Data Engineering Certificate:** Completed AWS data engineering program covering cloud data processing and analytics
- **Fuse Machines AI Fellowship:** Selected for a competitive fellowship program recognizing technical excellence and potential
- **IBM Data Engineering Specialization:** Completed IBM Data Engineering Specialization; acquired knowledge of SQL, ETL, pipelines, Airflow, Kafka, and cloud databases